# **Streamlining Ancestry**: A Lightweight Document Processing Pipeline for Historical Document Data Extraction

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Honors Program

## BACKGROUND

**Objective**: Automatic extraction of historical genealogical data

**Challenge**: Historical documents have many formats and individual handwriting can vary

**Focus**: Historical records, such as birth, marriage, and death certificates [1]

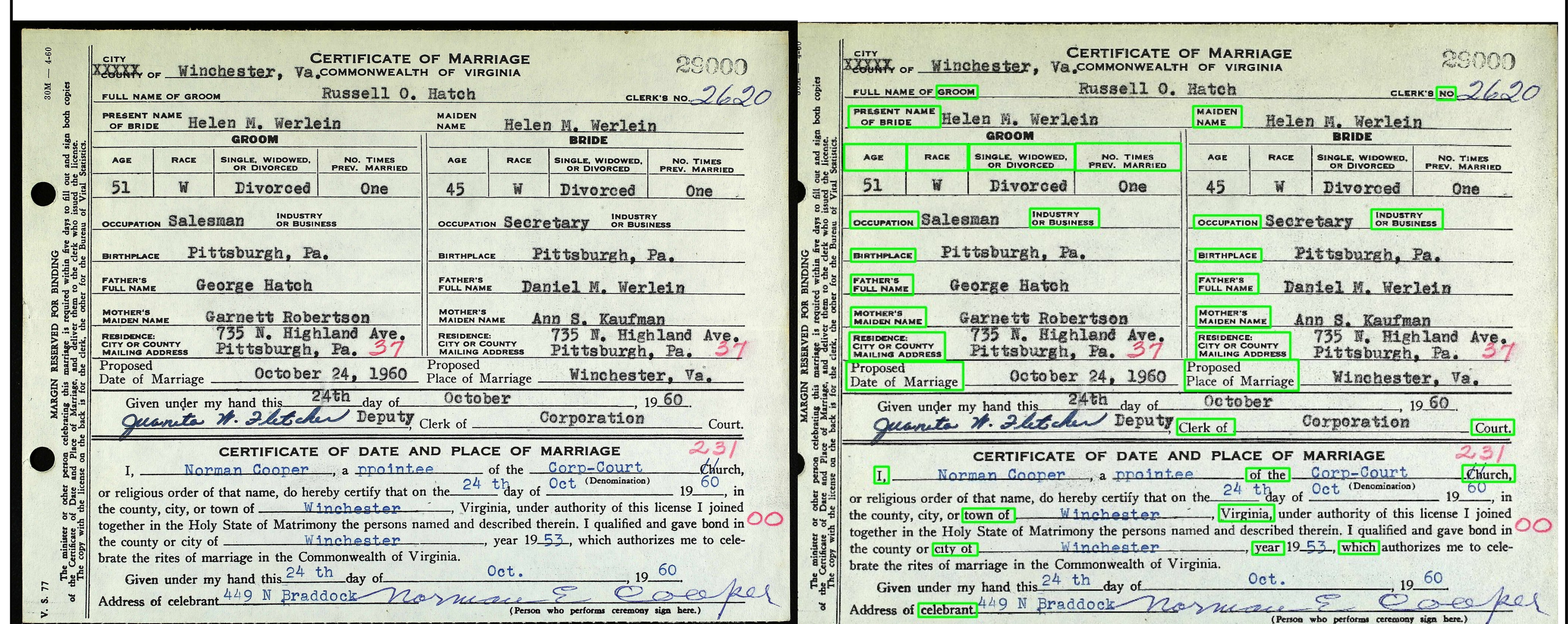

Fig 1. Unlabeled data (left) vs. labeled data (right)

## GOAL

Use **Object detection**[2] and **Optical Character Recognition**[3] to automate the process

**Simple**: Limit labeled data required

**Replicable**: Design should be easily replicable and transferable to similar problems

**Accuracy**: Aiming for over 90% accuracy for dependability

## DATASET

Sourced: Ancestry.com to label

Dataset Name: Virginia. Marriage Certificates October 1960

Size of dataset: 20 Images (15:5)

Augmentation: YOLOv8 defaults except for lr-flip and scale which were turned to 0.0

## METHODOLOGY

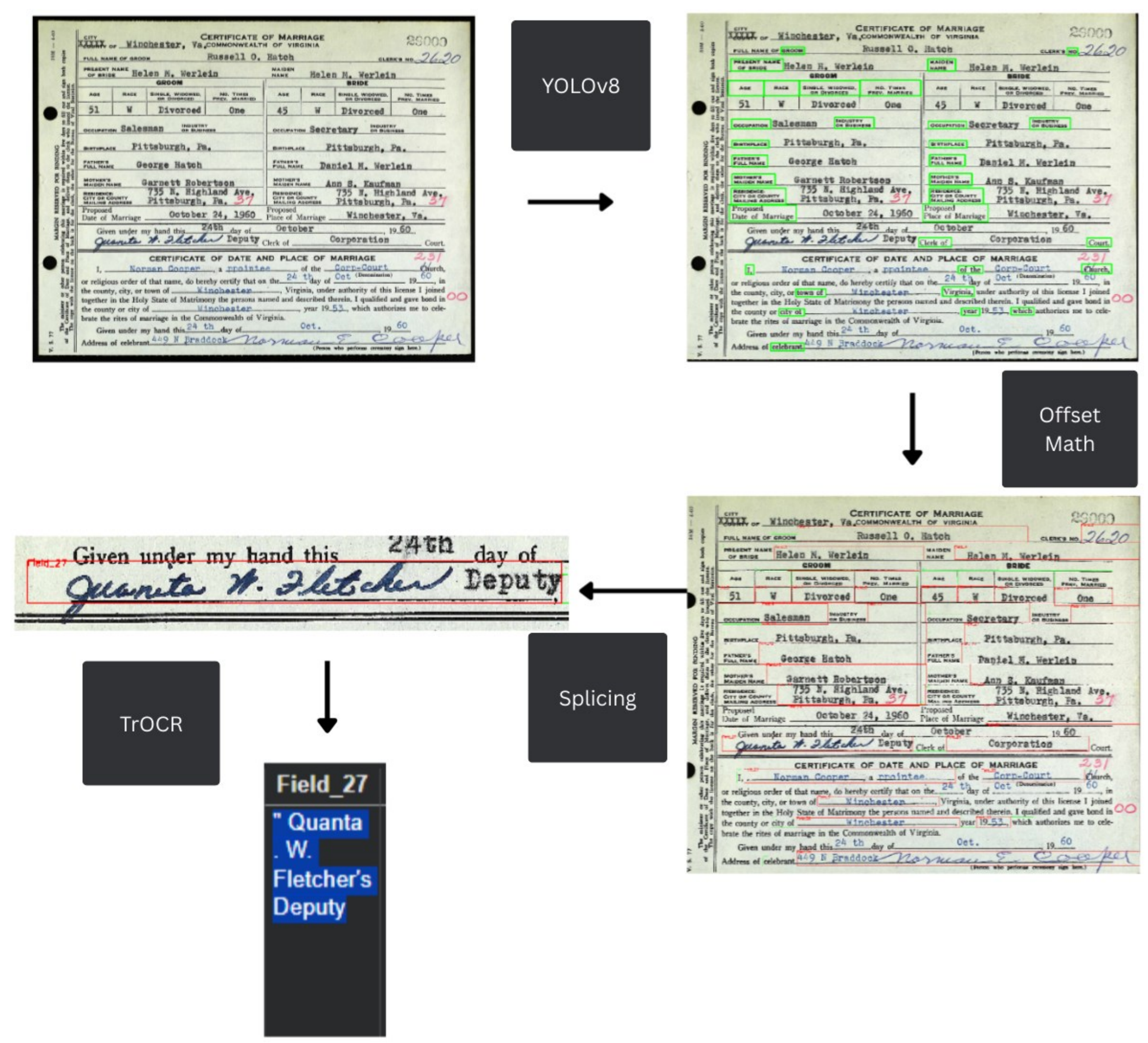


Fig 2. Journey our data will take through the system.

## RESULTS

After 200 epochs we have achieved 99.5% mAP and the following results:

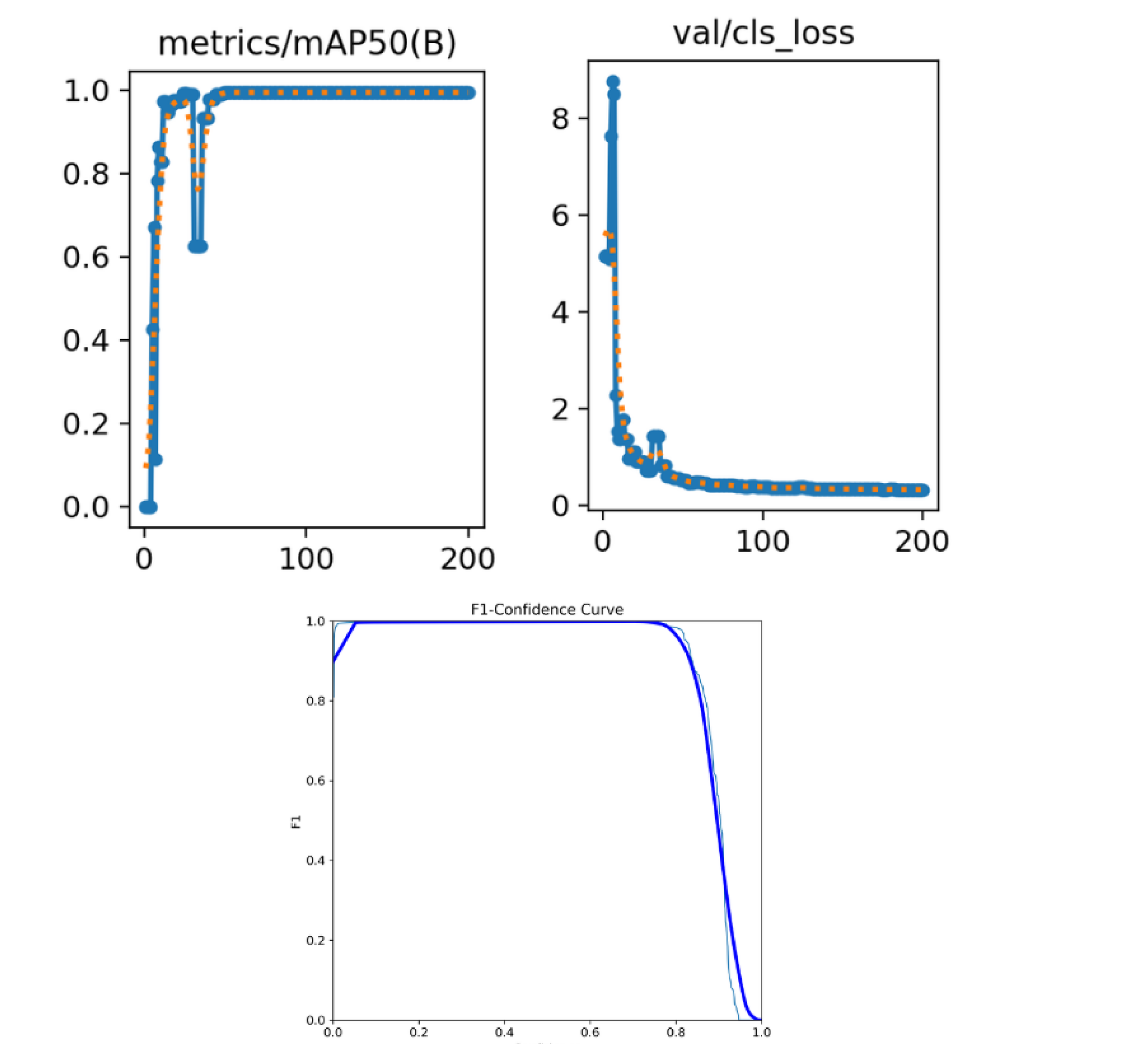


Fig 3. mAP50 graph (top left) class loss graph (top right) F-1 curve graph (bottom)

## FUTURE PLAN

- Greatly improve accuracy of OCR[3] models
- Improved ROI extraction with better equations to gather offset
- Test the same system on brand new dataset to confirm findings

## REFERENCES

1 Hoeve, C. D. (2018). Finding a place for genealogy and family history in the digital humanities. Digital Library Perspectives, 34(3), 215–226.

2 Martinez, J., Wortham, L., Majid, N., & Barney Smith, E. H. (2023). Extracting From Handwritten Historical Data from Registry Forms Using YOLOv8. Fort Lewis College / Luleå Technical University.

3 Jain, P., Taneja, K., & Taneja, H. (2021). Which OCR toolset is good and why? A comparative study. Kuwait Journal of Science, 48(2), 1–12.